LFM4Ads: Large Foundation Model for Ads Recommendation

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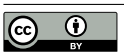
Abstract

Recent work leverages large-scale pretrained foundation models (FMs) to transfer generalizable representations to downstream advertising tasks. Most existing paradigms transfer user representations (URs) and item representations (IRs), while underutilizing the user-item cross representations (CRs) generated by deeper interaction layers. Although a recent approach transfers target-aware embeddings (TAEs), it requires real-time upstream FM inference alongside downstream model inference, introducing additional engineering complexity and overhead.

To address these limitations, we propose Large Foundation Model for Ads Recommendation (LFM4Ads), an efficient *pretrain-then-transfer* framework for cross-domain ad recommendation. LFM4Ads jointly exploits URs, IRs, and CRs from the upstream FM. It extracts sample-level CRs from pretrained interaction layers and aggregates them into user-level CRs using a time-aware exponential moving average, thereby decoupling upstream FM inference from downstream online serving. To use CRs efficiently, we introduce three downstream strategies at different levels of granularity: feature-level gating for CR fusion, module-level fine-tuning to yield downstream CRs, and model-level retrieval by matching CRs with IRs.

LFM4Ads has been deployed on Tencent's advertising platform since Q4 2024. It has supported 10+ launches across major scenarios, yielding a platform-wide gross merchandise value (GMV) lift of 2.45% and annual revenue gains in the hundreds of millions of dollars. Offline experiments on KuaiRand-1K further demonstrate the effectiveness of LFM4Ads. Our code is available at https://github.com/ZhuangQu/LFM4Ads.

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CCS Concepts

• **Information systems** → **Recommender systems**; **Online advertising**; • **Computing methodologies** → *Transfer learning*.

Keywords

Recommendation System, Pretraining, Transfer Learning

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1 Introduction

Online advertising has grown into a massive digital economy, driving hundreds of billions of dollars in annual revenue [13]. Its ranking and bidding systems depend on click-through rate (CTR) and conversion rate (CVR) prediction, motivating extensive research on multi-field feature interactions [5, 29, 35] and sequential behavior modeling [6, 50, 51]. As platforms span many scenarios and tasks, multi-task and multi-domain learning can share statistical strength [17, 24, 28, 38], but may suffer from negative transfer and the seesaw effect [1, 27, 40, 52]. Strict latency, throughput, and cost constraints also make a single large joint model difficult to deploy. This motivates the *pretrain-then-transfer* paradigm, in which foundation models (FMs) learn reusable knowledge from cross-scenario data and adapt it to specialized downstream models [4, 15, 18, 23, 32, 43, 49, 53].

However, existing FM transfer pipelines remain incomplete in both *what* is transferred and *how* it is utilized. First, most methods transfer user representations (URs) and item representations (IRs), typically obtained from upstream user or item towers [30, 32, 49]. Although URs and IRs can capture user-side and item-side characteristics, they underutilize finer-grained user-item correlations generated by deeper interaction layers. A recent work transfers Target-Aware Embeddings (TAEs) of user-item pairs [15], but it requires the FM to compute TAEs in real time, resulting in an additional upstream inference alongside downstream inference. Efficient transfer of user-item cross representations (CRs) therefore remains unresolved.

Second, the transferability of CRs is underexplored. Modern recommenders contain multiple explicit and implicit interaction layers [14, 39, 42], but it is unclear which layers preserve the most generalizable CRs. Moreover, CRs are sample-specific and lack reusable coarse-grained forms for downstream tasks. Additionally, existing transfer mechanisms are largely confined to feature-level injection through simple projections, which may fail to bridge upstream-downstream semantic gaps and may dilute FM signals among thousands of downstream features. Module-level reuse and model-level retrieval of FM representations are rarely explored.

To bridge these gaps, we propose Large Foundation Model for Ads Recommendation (LFM4Ads), a novel *pretrain-then-transfer* framework for cross-domain ad recommendation, as shown in Figure 1. Our contributions are:

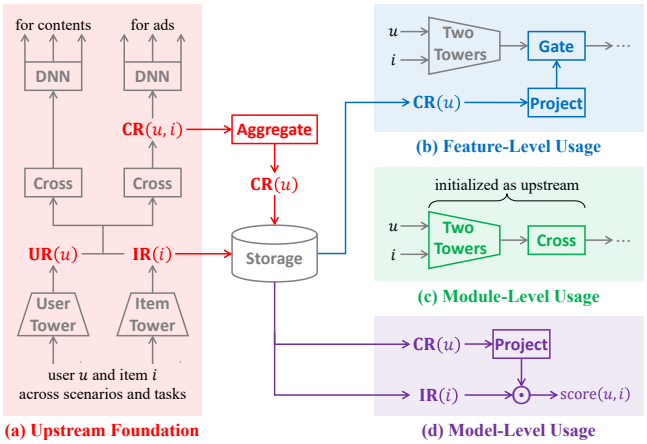


Figure 1: Overview. (a) LFM4Ads is pretrained on content and ad samples across scenarios and tasks, and representations are transferred to downstream models. (b) Cross representations are fused by gating for feature-level usage. (c) Modules that yield cross representations are fine-tuned for module-level usage. (d) Cross and item representations are matched for model-level usage.

- We extract all available representations from FM and leverage their complementary strengths: URs record user interests, IRs alleviate the item cold-start problem, and CRs provide finer-grained interaction signals that outperform isolated URs or IRs.
- We investigate CRs from diverse interaction layers and aggregate sample-level CRs into user-level CRs using a time-aware exponential moving average, making CRs reusable for downstream tasks.
- We transfer upstream knowledge at the feature, module, and model levels. Specifically, we fuse CRs via a nonlinear gate, reuse upstream modules that yield CRs, and perform retrieval directly using CRs and IRs. From feature-level to module-level and then to model-level, our transfer granularity increases and fewer parameters require training, demonstrating that *the stronger the foundation, the thinner the downstreams.*

We comprehensively evaluate LFM4Ads across eight scenarios in a public dataset, KuaiRand-1K. Our feature-level gating, module-level fine-tuning, and model-level retrieval improve the average AUC by 2.5‰, 1.5‰, and 18.9‰ over baselines, respectively.

LFM4Ads has supported 10+ production launches and yields a platform-wide gross merchandise value (GMV) lift of 2.45%, translating to annual revenue increases in the hundreds of millions of dollars. Specifically, feature-level fusion improves AUC by 1‰ with a 1.35% GMV lift in Weixin Moments. Module-level fine-tuning yields an overall GMV lift of 1.88% across multiple pCVR models. Model-level retrieval lifts GMV by 1.17% in Weixin Channels. Ablation studies further highlight the importance of CR aggregation and nonlinear gating.

2 Related Work

Industrial advertising and recommendation models learn user-item relevance using complementary techniques, including multi-field feature interaction modeling [5, 9, 42], multi-task learning [28], sequential behavior modeling [6, 50, 51], and multi-domain modeling [19, 20]. Together, they underpin modern click-through-rate and conversion-rate prediction pipelines. Although these models capture high-order interactions and evolving user interests, strict constraints on latency, computational resources, and feature budgets limit their scale in online serving [8, 11, 34].

To address these constraints, upstream-downstream systems pretrain large models on diverse data and transfer knowledge through representation injection or model adaptation. Representative systems learn sequence-based user embeddings [32, 49], transfer multimodal item signals [2, 36], or reuse learned user-item similarities [30]. Model-adaptation methods fine-tune pretrained models of user activity sequences [4] or distill knowledge from an external foundation model into smaller domain-specific models [23]. Despite these advances, sample-specific interaction knowledge from deeper layers remains difficult to store, transfer, and reuse across heterogeneous downstream architectures.

A parallel line of research explores large language model (LLM)-based approaches to e-commerce instruction tuning [22, 33, 37] and generative, cross-domain, or generalizable recommendation [43, 46, 53]. Collectively, these studies explore language-based and generative methods for e-commerce and recommendation. Recent work also studies dimensional degeneration in generative ranking models with LLM-inspired architectures [45].

3 Methodology

We first design our upstream foundation model in Section 3.1, then enhance cross representations in Section 3.2, and finally discuss downstream usages in Section 3.3.

3.1 Pretrain and Extract Representations

Our Large Foundation Model for Ads Recommendation (LFM4Ads) is illustrated in Figure 1a. We adopt a *two-tower* design, where a user tower extracts a user representation (UR) from user features, and an item tower extracts an item representation (IR) from item features. Then we employ interaction layers (e.g., DCNv2 [42]) to capture the user-item cross correlations.

Formally, let $\mathbf{UR}(u)$ be the representation of user u, and $\mathbf{IR}(i)$ be the representation of item i. Whenever a sample (u, i) arrives at LFM4Ads, we extract $\mathbf{UR}(u)$ and $\mathbf{IR}(i)$ from the two towers. Besides, the interaction layers capture the correlations between $\mathbf{UR}(u)$ and $\mathbf{IR}(i)$ at finer-grained levels, including explicit and implicit feature interactions. To leverage these interactions, we extract the output of an interaction layer as a user-item cross representation (CR), denoted as $\mathbf{CR}(u, i)$. *Together, URs, IRs, and CRs constitute the three representation types transferred in LFM4Ads.*

LFM4Ads is pretrained on all samples from multiple scenarios (e.g., Weixin Moments, Search Ads), multiple tasks (e.g., pCTR, pCVR), and two domains (content and advertising). To account for distributional differences between content and ad samples, we adopt a *dual-branch* design after the two towers, with one branch for content tasks and the other for ad tasks. Notably, we transfer CRs only from the ad branch because our downstream tasks involve ad recommendation. For content tasks, transferring the CRs from the content branch is also feasible.

3.2 Enhance Transferability of CRs

There are two main challenges when transferring CRs: (1) CR is a sample-level representation of a user-item pair, which is too fine-grained for downstream tasks. (2) It is impractical to calculate and store the CRs for all user-item pairs. Although a recent approach [15] computes Target-Aware Embeddings (TAEs) in real time, it requires an additional upstream inference pass alongside downstream inference.

To address these challenges, we aggregate sample-level CRs for each user into a single user-level CR. Formally, let $\mathbf{CR}(u)$ be the aggregated CR of user u. We initialize $\mathbf{CR}(u) \leftarrow \mathbf{0}$. Whenever a new $\mathbf{CR}(u, i)$ is generated, we update $\mathbf{CR}(u)$ by

$$\mathbf{CR}(u) \leftarrow \beta(t) \cdot \mathbf{CR}(u) + \big(1 - \beta(t)\big) \cdot \mathbf{CR}(u, i), \tag{1}$$

where t is the time interval since the previous update of $\mathbf{CR}(u)$, and $\beta : \mathbb{R}_+ \to [0, 1]$ is a decreasing function.

We refer to (1) as a *time-aware exponential moving average*, which adapts to user activity. For inactive users, t is large and we have $\beta(t) \approx 0$ to prioritize current interactions and address distribution drift. For active users, t is small and we have $\beta(t) \approx 1$ to retain more historical interactions.

To further enhance the transferability of CRs, we explore which layer yields the best CRs. We find that the optimal CRs are extracted from the middle layer for feature-level usage, and from the deepest layer for model-level usage. More details are in Sections 4.2 and 4.4.

3.3 Three Downstream Usages

In this section, we discuss how to use CRs for downstream tasks. We propose feature-level, module-level, and model-level usages, ranging from fine-grained to coarse-grained.

3.3.1 Feature-Level Usage: Nonlinear Gating. A straightforward fusion strategy is to treat an upstream representation as an additional feature and concatenate it with downstream embeddings, as in SUM [49]. Formally, let $\boldsymbol{E}$ be the concatenation of downstream embeddings. The concat-based fusion yields

$$\boldsymbol{E}' = \big[\boldsymbol{E}, \quad \boldsymbol{M} \cdot \mathbf{CR}(u)\big], \tag{2}$$

where $\boldsymbol{M}$ is a learnable projection matrix that reduces the dimensionality of $\mathbf{CR}(u)$.

However, (2) may be ineffective because it directly combines representations from two semantically misaligned spaces. We tried several approaches, including employing an alignment loss between the upstream and downstream features [21], but none of them led to a performance gain over production baselines during online A/B tests. Inspired by self-gating [3, 12, 47], we adopt a gating-based fusion, which yields

$$\boldsymbol{E}' = \boldsymbol{E} \odot \boldsymbol{\sigma}\big(\boldsymbol{M} \cdot \mathbf{CR}(u)\big), \tag{3}$$

where $\boldsymbol{\sigma}$ is a nonlinear function and $\odot$ denotes the Hadamard product. $\boldsymbol{M}$ projects $\mathbf{CR}(u)$ to the same dimensionality as $\boldsymbol{E}$. Here $\boldsymbol{\sigma}(\boldsymbol{M}\cdot)$ serves as a gate as illustrated in Figure 1b. A recent work [47] revealed the mechanism behind (3) — $\boldsymbol{\sigma}(\boldsymbol{M}\cdot)$ *serves as a nonlinear*

encoder to construct new representations, thereby mitigating dimensional collapse and representation redundancy.

3.3.2 Module-Level Usage: Fine-Tuning Cross Modules. As discussed in Section 3.2, there are challenges in transferring upstream $\mathbf{CR}(u, i)$. To overcome this, we proposed transferring the aggregated $\mathbf{CR}(u)$ in Section 3.2. In this section, we propose another solution — transferring the upstream modules that yield $\mathbf{CR}(u, i)$.

As illustrated in Figure 1c, we initialize the downstream model with the same modules as LFM4Ads, including the two towers and the interaction layers in the ad branch. Thus, the downstream model yields the same $\mathbf{CR}(u, i)$ as LFM4Ads. We then fine-tune these modules on downstream samples to obtain $\mathbf{CR}'(u, i)$, aiming to better align the cross representations with downstream requirements.

Inspired by prior work showing that pretrained representations can transfer learned user-item similarity structures across models [30, 31, 36], we transfer these upstream modules together with the user-item correlations (i.e., distance or similarity) learned by LFM4Ads. This makes downstream learning start from a representation space where user-item interactions have already been broadly shaped by multi-scenario pretraining.

3.3.3 Model-Level Usage: Standalone Retrieval. URs and IRs learned by LFM4Ads naturally encode the relevance between users and items, so they can be directly used for retrieval [32]. Formally, given a user u, we retrieve the best-matching item i from the candidate pool, with the highest

$$\text{score}(u, i) = \left\langle \mathbf{Adapter}\big(\mathbf{UR}(u)\big), \quad \mathbf{IR}(i) \right\rangle, \tag{4}$$

where $\langle \cdot, \cdot \rangle$ denotes the dot product, and **Adapter** is a tiny learnable layer (e.g., a projection matrix). We optimize **Adapter** using binary cross-entropy (BCE) or the InfoNCE [41].

Since our $\mathbf{CR}(u)$ can be regarded as a user representation like $\mathbf{UR}(u)$, we explore replacing $\mathbf{UR}(u)$ with $\mathbf{CR}(u)$ in (4), and find that $\mathbf{CR}(u)$ generally performs better. More details are in Section 4.4.

As shown in Figure 1d, our retrieval model (4) does not require parameter-heavy two towers. Benefiting from our effective and refined $\mathbf{CR}(u)$, a tiny adapter is sufficient to serve as a whole retrieval model. This highlights the potential of LFM4Ads — *it provides not only an additional representation, but also a standalone model.*

3.3.4 Trade-off among Three Usages. Feature-level usage is compatible with existing online ranking models, since it only injects an additional representation as a new feature. It can provide incremental performance gains for well-trained mature systems.

Module-level usage is suitable for new ranking models. It not only accelerates training convergence but also leverages full-scenario knowledge to tackle long-tail and sparse scenarios. But it lacks flexibility, since it requires upstream and downstream models to share similar architectures.

Model-level usage is convenient for fast deployment. With a small number of parameters and low latency, it is suitable for large-scale retrieval. However, model-level usage achieves lower AUC than the other two usages, possibly due to its more limited inputs and smaller model capacity.

From feature-level to module-level and then to model-level, *the transfer granularity ranges from fine to coarse, and thus the downstream training cost shifts from heavy to light.*

4 Offline Experiments

There is no widely adopted benchmark for evaluating recommendation foundation models. We therefore use KuaiRand-1K [7], which contains interactions spanning 15 scenarios, to emulate a *pretrain–then-transfer* pipeline. Specifically, we pretrain an upstream foundation model on the union of data from all scenarios, and then transfer it to 15 scenario-specific downstream models, each trained only on samples from its own scenario.

4.1 Settings

4.1.1 Dataset. KuaiRand-1K contains 1,000 users, 4,369,953 items, and 11,713,045 samples. These samples cover 15 scenarios and span 31 days. We use samples from the first 25 days for training, the following 3 days for validation, and the last 3 days for testing. Each downstream dataset contains only samples from a single scenario. We discard continuous features and retain 27 user-side and 9 item-side categorical features for each sample. We take click or not as the label, constructing a pCTR task.

4.1.2 Upstream Model. Our foundation model comprises a user tower, an item tower, and a DCNv2 [42]. The DCNv2 has 3 cross layers and a 3-layer DNN. All layers are sequentially denoted as

$$Towers,\ Cross_1,\ Cross_2,\ Cross_3,\ DNN_1,\ DNN_2,\ DNN_3.$$

$Towers$ map each feature to an embedding of length 10, and then concatenate them. The size of $Cross_1, \cdots, DNN_2$ is 360×360, while the size of DNN_3 is 360×15. The foundation model consumes samples from all 15 scenarios, emulating online cross-scenario pretraining. The output is 15 logits for 15 scenarios, but only the logit for the sample's scenario is used.

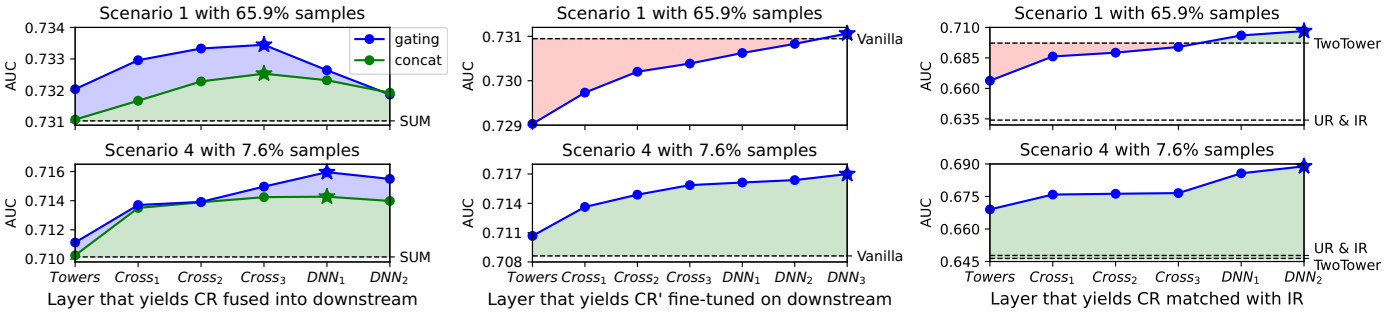
4.1.3 Training. Our foundation and downstream models share the same training settings. The loss is BCE and the optimizer is AdamW [26]. The settings of AdamW follow PyTorch defaults, where the learning rate is 0.001, betas are (0.9, 0.999), and weight decay is 0.01. The samples are sorted by timestamp and the batch size is 10,000. We stop training when the validation AUC declines.

4.1.4 Upstream Representations. After pretraining, we freeze the foundation model and obtain $\mathbf{UR}(u)$ and $\mathbf{CR}(u)$ for each user u, and $\mathbf{IR}(i)$ for each item i. $\mathbf{UR}(u)$ and $\mathbf{IR}(i)$ are extracted from $Towers$. To obtain $\mathbf{CR}(u)$, we replay the training and validation samples. For each sample (u, i), we extract $\mathbf{CR}(u, i)$ from $Towers$, $Cross_1$, $\cdots$, or DNN_2, then aggregate sample-level $\mathbf{CR}(u, i)$ into user-level $\mathbf{CR}(u)$ using (1) with $\beta = 0.99$.

4.1.5 Evaluation. We evaluate the test AUC of each downstream model, which is trained only on samples from a specific scenario. We average the test AUC over 100 runs, each involving a foundation model and its corresponding downstream model.

4.2 Feature-Level Usage

We train downstream models with the same architecture as the upstream FM and fuse $\mathbf{CR}(u)$ into the downstream embeddings. For concat-based fusion (2), the dimensionality of $\mathbf{CR}(u)$ is reduced to 10 to match a downstream embedding. For gating-based fusion (3), we let $\boldsymbol{\sigma} = \mathbf{sigmoid}$ to introduce nonlinearity. As a baseline, SUM fuses $\mathbf{UR}(u)$ via concat [49].



(a) Feature-level usage. SUM fuses UR via concat [49]. ■ indicates the gain from CR, and ■ indicates the gain from gating.

(b) Module-level usage. Vanilla trains from scratch. ■ indicates the gain from upstream initialization, while ■ indicates the drop.

(c) Model-level usage. TwoTower trains its two towers. ■ indicates the gain from CR & IR, while ■ indicates the drop.

Figure 2: Offline results on two typical scenarios of KuaiRand-1K.

Figure 2a compares feature-level usage in Scenario 1, which is a typical large scenario, and Scenario 4, which is a typical medium-sized scenario. Concatenating CRs generally outperforms SUM, as indicated by ■, and gating with CRs further improves AUC, as indicated by ■. With CRs extracted from $Cross_3$, concat (2) improves AUC over SUM by 1.5‰ and 4.1‰ in Scenarios 1 and 4, respectively. Gating (3) yields additional gains of 0.9‰ and 0.8‰ over concat. Even a 1‰ AUC gain can be practically meaningful in recommendation [16, 25]. These gains suggest that CRs provide useful information about finer-grained user-item interactions and that gating helps downstream models leverage this information. Gating (3) fuses CRs into downstream embeddings without increasing their dimensionality.

In both scenarios, AUC first increases and then decreases as the layer for extracting CRs becomes deeper. The highest AUC is achieved at $Cross_3$ in Scenario 1 and DNN_1 in Scenario 4, both of which are intermediate layers. This pattern suggests a balance between representation refinement and task specificity, consistent with [48]. Shallow CRs may be insufficiently processed for downstream use, whereas deeper CRs may be too specialized to upstream tasks. Intermediate CRs provide useful interaction information while retaining flexibility for downstream models.

4.3 Module-Level Usage

We train downstream models with the same architecture as the upstream FM. The first several layers are initialized with upstream weights, while the remaining layers are initialized from scratch. As a baseline, Vanilla initializes all layers from scratch.

Figure 2b shows that AUC increases monotonically as more layers are transferred, with the highest AUC achieved when all layers are transferred in both scenarios. This trend suggests that transferring more layers provides more of the user-item interaction knowledge learned by the FM.

However, the gains over Vanilla also vary with scenario size. In Scenario 1 with 65.9% samples, partial transfer leads to negative transfer, as indicated by ■. In Scenario 4 with 7.6% samples, all evaluated transfer depths outperform Vanilla, as indicated by ■. Our explanation is that, when downstream samples are abundant, the downstream model can learn effectively from scratch. Partial transfer introduces the bias from the FM, but subsequent layers do not correct this bias, resulting in negative transfer.

4.4 Model-Level Usage

We instantiate the **Adapter** in (4) as a learnable matrix $\boldsymbol{W}$, i.e.,

$$\text{score}(u, i) = \mathbf{UR}^T(u) \cdot \boldsymbol{W} \cdot \mathbf{IR}(i). \tag{5}$$

We also evaluate a variant that replaces URs with CRs:

$$\text{score}(u, i) = \mathbf{CR}^T(u) \cdot \boldsymbol{W} \cdot \mathbf{IR}(i). \tag{6}$$

As a baseline, we train a model with its own two towers, formalized as

$$\text{score}(u, i) = \mathbf{UserTower}^T(u) \cdot \boldsymbol{W} \cdot \mathbf{ItemTower}(i), \tag{7}$$

where **UserTower** and **ItemTower** have the same architecture as FM's, and are trained from scratch on downstream samples. We denote Model (5) as UR & IR, Model (6) as CR & IR, and Model (7) as TwoTower, respectively.

Figure 2c compares the three retrieval models. TwoTower outperforms UR & IR in Scenario 1, while UR & IR slightly outperforms TwoTower in Scenario 4. With CRs extracted from DNN_2, CR & IR outperforms both baselines, as indicated by ■. Its AUC gains over TwoTower are 9.8‰ and 42.5‰ in Scenarios 1 and 4, respectively, and those over UR & IR are 73.0‰ and 41.1‰. These results suggest that abundant downstream samples support learning effective towers from scratch, whereas upstream representations become more useful when downstream supervision is limited. The further gains from CRs suggest that the user-item interactions captured upstream provide useful information beyond the tower representations alone.

In both scenarios, AUC increases monotonically as the layer for extracting CRs becomes deeper and peaks at DNN_2, the deepest layer for extracting CRs. This pattern may arise because CR & IR trains only a matrix adapter, whose limited capacity makes the model more dependent on the refined knowledge provided by deeper upstream layers. Reusing these upstream CRs and IRs avoids the need to train and store downstream two towers, leaving only a small adapter to be trained for retrieval.

Table 1: Offline results on all available scenarios of KuaiRand-1K. Highest is bolded and lowest is italicized. Average is weighted by the number of samples. CR is extracted from $Cross_3$ for feature-level usage, and DNN_2 for model-level usage. All layers are fine-tuned to yield CR′ for module-level usage.

Scenario		1	0	4	2	6	3	8	5	Average
# of samples		7,717,601	2,407,352	895,385	402,293	183,403	37,418	17,430	16,424	-
% of samples		65.9%	20.6%	7.6%	3.4%	1.6%	0.3%	0.1%	0.1%	-
Feature	**SUM**	*0.7310*	0.7123	*0.7101*	*0.7975*	*0.7183*	*0.7017*	*0.7622*	*0.6562*	*0.7275*
	concat CR	0.7325	0.7123	0.7142	0.7993	**0.7277**	0.7114	**0.7668**	0.6622	0.7291
	gating CR	**0.7334**	**0.7132**	**0.7150**	**0.8000**	0.7249	**0.7302**	0.7657	**0.6699**	**0.7300**
Module	**Vanilla**	0.7309	0.7120	0.7086	0.7971	0.7173	0.7038	0.7627	0.6606	0.7273
	CR′	**0.7311**	**0.7136**	**0.7170**	**0.7990**	**0.7361**	**0.7387**	**0.7861**	**0.6834**	**0.7288**
Model	**TwoTower**	0.6972	0.6450	*0.6464*	*0.7570*	*0.5776*	*0.5838*	**0.6550**	*0.5562*	0.6821
	UR & IR	*0.6340*	*0.6376*	0.6478	**0.7662**	0.5915	0.5911	0.6449	0.5565	*0.6395*
	CR & IR	**0.7070**	**0.6821**	**0.6889**	0.7645	**0.6553**	**0.6122**	*0.5646*	**0.6012**	**0.7010**

4.5 Overall Comparisons

We conduct comprehensive experiments on the eight scenarios with sufficient test samples in KuaiRand-1K. Based on the preceding experiments, we adopt the following best practices: extracting CRs from $Cross_3$ for feature-level usage, initializing all layers with upstream weights for module-level usage, and extracting CRs from DNN_2 for model-level usage.

Table 1 shows that all three usages generally improve AUC over their respective baselines. Feature-level gating yields an average AUC gain of 2.5‰ over SUM, module-level fine-tuning yields 1.5‰ over Vanilla, and model-level retrieval yields 18.9‰ over TwoTower.

Note that the gains tend to be larger in smaller downstream scenarios. In Scenarios 6, 3, 8, and 5, the four smallest scenarios, the largest AUC gains span 23.4–77.7‰ over the corresponding baselines.

5 Online Deployment

We instantiate LFM4Ads as an industrial system, and deploy it on Tencent's advertising platform at scale.

5.1 Samples and Features

We collect samples from all available advertising scenarios, including Weixin Moments, Weixin Channels, Weixin Official Accounts, Tencent News, Tencent Sports, Demand-Side Platform, and other pCTR and pCVR scenarios. Besides advertisements, we also collect samples from the content domain, which is on a much larger scale and hence contains more collaborative information about users' interests. The total number of samples fed into LFM4Ads is tens of billions per day, with 80% from the content domain and 20% from the ad domain. The number of samples is 20 to hundreds of times larger than that of individual downstream tasks.

Each upstream sample contains approximately 1,800 features, most of which are on the user side. In contrast, most downstream ranking samples have 200–1,000 features, while retrieval and matching samples have 100–500 features. 87% of the user features are multi-valued, including around 50 sequences, with lengths ranging from dozens to thousands. On average, flattening all multi-valued features yields more than 60,000 feature values per sample.

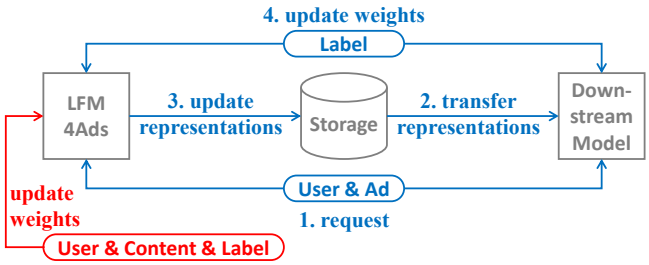


Figure 3: Online response process.

5.2 Scale of Models

LFM4Ads has a 4-TB parameter footprint, which is 48% larger than the largest downstream model, Weixin Moments pCTR, 233% larger than the Weixin Channels pCTR model, and 427% larger than the average size of downstream models. It processes 6.3B sparse features, which is 270% larger than the largest downstream task's. The FLOPS is 1.45B and the QPS is 500K. To save storage, we employ quantization to compress an upstream representation from 4KB down to 1KB.

5.3 Response Process

The response process of our online system is shown in Figure 3. Formally, whenever a user u and an ad i arrive, we retrieve $\mathbf{UR}(u)$, $\mathbf{IR}(i)$, and $\mathbf{CR}(u)$ from the storage, and transfer them into the downstream model. Meanwhile, LFM4Ads performs inference on (u, i) and extracts new $\mathbf{UR}(u)$, $\mathbf{IR}(i)$ and $\mathbf{CR}(u, i)$. Then in the storage, $\mathbf{UR}(u)$ and $\mathbf{IR}(i)$ are updated directly, and $\mathbf{CR}(u)$ is aggregated by (1). Whenever the label arrives (i.e., the user reacts to the ad), we update the weights of LFM4Ads and the downstream model. In addition, LFM4Ads is updated irregularly by content samples.

Table 2: Online A/B results.

Usage	Scenario	Task	GMV
Feature	Weixin Moments	pCTR	+0.42%
	Weixin Moments	pCVR	+2.53%
	Weixin Channels	matching	+0.70%
	Search Ads	matching	+1.75%
	Demand-Side Platform	pCTR	+0.76%
	Internet Service	pLTV	+0.93%
Module	Weixin Moments	pCVR	+0.92%
	Weixin Channels	pCVR	+0.92%
	Weixin Official Accounts	pCVR	+0.94%
	Content Platform	pCVR	+1.59%
	Audience Network	pCVR	+0.95%
Model	Weixin Channels	retrieval	+1.17%

Table 3: Online ablations on transfer methods.

Transferred	Branch	Fusion	AUC	GMV
None	-	-	0.8515	-
$\mathbf{UR}(u)$	single	concat	0.8501	negative
$\mathbf{UR}(u)$	dual	concat	0.8520	+0.79%
$\mathbf{CR}(u), \beta = 0$	dual	gating	0.8513	negative
$\mathbf{CR}(u), \boldsymbol{\sigma} = \mathbf{id}$	dual	gating	0.8521	negative
$\mathbf{CR}(u)$	dual	gating	**0.8525**	**+1.35%**

Table 4: Online ablations on upstream domains.

	Static Only	+ Content Sequence	+ Ad Sequence	+ Both Sequences
AUC gain	-	+0.031	+0.042	+0.052

5.4 Launches and Productization

LFM4Ads has been productized in 10+ major online launches, lifting overall GMV by 2.45% across the platform. At the scenario level, Weixin Moments and Channels achieve GMV lifts of 2.16% and 1.18%, respectively. At the task level, pCTR, pCVR, and pLTV models bring improvements of 0.18%, 1.33%, and 0.44%, respectively.

Table 2 lists the online A/B test results of the three usages. Feature-level usage generally improves diverse scenarios and tasks, with the largest GMV gain of +2.53% on the Weixin Moments pCVR model, indicating substantial business impact. For module-level usage, LFM4Ads modules have been productized in multiple pCVR models, contributing to an overall GMV lift of 1.88%. In addition to the GMV gain, it also speeds up downstream convergence and reduces training costs. For model-level usage, a standalone retrieval model achieves a +1.17% GMV lift in Weixin Channels, while also improving CTR by 1.83%, CTCVR by 3.34%, reducing quick-slip rate by 0.36%, and increasing average dwell time by 1.66%.

5.5 Online Ablation

Table 3 reports online ablations in Weixin Moments. Single-branch SUM degrades downstream performance, whereas dual-branch SUM yields a 0.79% GMV lift, highlighting the importance of separating content and ad branches. LFM4Ads further increases the GMV lift to 1.35%, demonstrating the benefit of CR transfer with nonlinear gating.

Setting $\beta = 0$ in (1) reduces AUC relative to full LFM4Ads and turns the GMV lift negative, highlighting the importance of aggregating historical interactions. Replacing the nonlinear gating function in (3) with the identity function $\boldsymbol{\sigma} = \mathbf{id}$ also reduces AUC and yields a negative GMV lift, showing the importance of nonlinearity in CR fusion.

5.6 Cross-Domain Analysis

We examine how upstream behavior sequences from the content and ad domains contribute to downstream ad recommendation. The upstream baseline relies solely on non-sequential features that describe users' static attributes. The upstream representations are transferred to the Weixin Channels pCTR model for ad recommendation.

As shown in Table 4, incorporating behavior sequences from both the content and ad domains yields the largest AUC gain of 52‰. This exceeds the gain from ad-domain sequences alone by 10‰ and that from content-domain sequences alone by 21‰. These results suggest that content-domain behavior provides complementary information for ad prediction when combined with ad-domain behavior.

6 Conclusion

LFM4Ads is a pretrain-then-transfer framework for cross-domain ad recommendation, transferring upstream user, item, and cross representations at multiple granularities. Feature-level gating, module-level fine-tuning, and model-level retrieval accommodate different downstream architectures and deployment requirements. Offline experiments on KuaiRand-1K and production deployment on Tencent's advertising platform validate its effectiveness. Extending transferable recommendation representations to downstream multi-channel bidding with explicit budget and ROI constraints remains an important direction [10, 44].

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GenAI Usage Disclosure

Generative AI assisted only with manuscript language, grammar, phrasing, and minor formatting. It did not generate research ideas, design methods, process or analyze data, conduct experiments, write or execute code, or produce results. All technical content, experimental results, claims, and conclusions were independently developed, reviewed, and verified by the authors.

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